Numerical Solutions to Problems in Electrochemistry

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This thesis is dedicated to
someone
for some special reason

Abstract

This is a project on PINNs that do not seem to work very well.

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Chapter 1

Introduction

Numerical Electrochemisty is the study of the etc. Nikil is a nonce.

Chapter 2

Electrochemical Systems

Electrochemists study chemical reactions that involve electron transfer for many reasons such as biochemical sensing, electroplating, energy storage, imaging etc. A popular technique to understand the diffusive or absortive nature, thermodynamic parameters, as well as the existence of coupled homogenous chemical reactions is cyclic voltammetry.

2.1 Mathematical model of an electrochemical system

We can represent a general reduction-oxidation (redox) process at the electrode as:

$$\mathrm{A} + \mathrm{e}^- \rightleftharpoons \mathrm{B} \tag{2.1}$$

In a typical experiment, the potential is swept from some starting potential, E_i where the species A is not electroreduced, to some – more negative – vertex potential, E_v, at which the electron transfer between the species A and the electrode is rapid and the species B is formed. The potential is then swept back to E_i causing electron transfer in the opposite direction and thus the reformation of A. Throughout this process the current, I, is recorded. Plotting the current against the potential gives the characteristic cyclic voltammagram in

$$\frac{\partial \mathrm{C}(X,T)}{\partial T} = \frac{\partial^2 C(X,T)}{\partial X^2} \tag{2.2}$$

Subject to:

$$\begin{aligned} C(X,0) &= 1 \\ \left.\frac{\partial C(X,T)}{\partial X}\right|_{X=0} &= K_0\big(C(0,T)e^{-\alpha\theta} - \big(1 - C(0,T)\big)e^{(1-\alpha)\theta}\big) \end{aligned} \tag{2.3}$$